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### Carton and blank therefor

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#### Abstract

Aspects of the disclosure relate to a carrier for packaging one or more articles. The carrier is formed from a set of connected blanks each of which form a top engaging carrier, wherein the blanks are detachably connected together in a matrix fashion wherein each blank has a plurality of top-receiving apertures wherein a first distance (D.sub.1) between the respective centres of two adjacent ones of the apertures of a first blank is less than a second distance (D.sub.2) between the centre of a corner aperture of the first blank and the centre of the adjacent corner aperture of a second blank and wherein the first distance (D.sub.1) is less than a third distance (D.sub.3) between the centre of the corner aperture of the first blank and the centre of the adjacent corner aperture of a third blank.

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## Background/Summary

### TECHNICAL FIELD

(1) The present invention relates to cartons and to blanks for forming the same. More specifically, but not exclusively, the invention relates to a plurality of connected blanks for forming carriers of the top-gripping type having one or more apertures for receiving and retaining an article therein.

### BACKGROUND

(2) In the field of packaging it is known to provide cartons for carrying multiple articles. Cartons are well known in the art and are useful for enabling consumers to transport, store and access a group of articles for consumption. For cost and environmental considerations, such cartons or carriers need to be formed from as little material as possible and cause as little wastage in the materials from which they are formed as possible. Further considerations are the strength of the carton and its suitability for holding and transporting large weights of articles. It is desirable that

the contents of the carton are secure within the carton.

(3) It is well known to provide top gripping article carriers in which an aperture is formed in a panel of the carrier, wherein tabs are struck from said aperture. The tabs are displaced out of the plane of said panel when an article is received in the aperture, wherein said tabs engage the article generally about a flange or lip of the article.

(4) It is desirable to simplify construction of the blank into a carrier, it is an object of the present disclosure to enable a plurality of connected blanks to be simultaneously assembled into packages. The carrier should be sufficiently robust to withstand the load of the articles. It is desirable that the carriers should be capable of packaging generally cylindrical articles, such as but not limited to beverage cans, which articles are of a sleek or slim design.

(5) The present invention seeks to provide an improvement in the field of cartons, typically formed from paperboard or the like.

#### SUMMARY

(6) A first aspect of the invention provides a set of connected blanks each for forming a top engaging carrier, wherein the blanks are detachably connected together in a matrix fashion wherein each blank has a plurality of top-receiving apertures wherein a first distance  $D_{sub.1}$  between the respective centres of two adjacent ones of the apertures of a first blank is less than a second distance  $D_{sub.2}$  between the centre of a corner aperture of the first blank and a centre of an adjacent corner aperture of a second blank and wherein the first distance  $D_{sub.1}$  is less than a third distance  $D_{sub.3}$  between the centre of the corner aperture of the first blank and a centre of an adjacent corner aperture of a third blank.

(7) Optionally, the first distance  $D_{sub.1}$  is generally equal to the maximum diameter  $D_{sub.c}$  of a generally cylindrical article adapted to be engaged by the first blank.

(8) Optionally, a corner radius  $R_{sub.b}$  of each blank is greater than a half of the maximum diameter  $D_c$  of a generally cylindrical article adapted to be engaged by that blank.

(9) Optionally, the corner radius  $R_{sub.b}$  is at least  $\frac{1}{8}$  inch (3.175 mm) greater than a half of the maximum diameter  $D_{sub.c}$  of a generally cylindrical article adapted to be engaged by that blank.

(10) Optionally, the second distance  $D_{sub.2}$  is generally equal to the third distance  $D_{sub.3}$ .

(11) Optionally, each of the second and third distances  $D_{sub.2}$ ,  $D_{sub.3}$  is generally equal to twice a corner radius  $R_{sub.b}$  of each blank.

(12) Optionally, a fourth distance  $R_{sub.s}$  is defined between the outer end of each tab-defining element and the centre of the respective aperture, the fourth distance  $R_{sub.s}$  is equal to or less than a half of the maximum diameter  $D_{sub.c}$  of a generally cylindrical article adapted to be engaged by that blank.

(13) Optionally, any point on the outline of a finger aperture is disposed outside an area defined by a radius equal to half of the maximum diameter ( $\frac{1}{2}D_{sub.c}$ ) around the centre of any one of the apertures which surround the finger aperture.

(14) A second aspect of the invention provides a blank for forming a top engaging carrier, the blank comprising a plurality of apertures each for receiving a portion of a generally cylindrical article wherein a first distance  $D_{sub.1}$  between the centres of two adjacent ones of the apertures is generally equal to the maximum diameter  $D_{sub.c}$  of the article and a corner radius  $R_{sub.b}$  of the blank is greater than a half of the maximum diameter  $D_{sub.c}$  of the article.

(15) Optionally, the corner radius  $R_{sub.b}$  is at least  $\frac{1}{8}$  inch (3.175 mm) greater than a half of the maximum diameter  $D_{sub.c}$  of the article.

(16) Optionally, a fourth distance  $R_{sub.s}$  is defined between the outer end of each tab-defining element and the centre of the respective aperture, the fourth distance  $R_{sub.s}$  is equal to or less than a half of the maximum diameter  $D_{sub.c}$  of a generally cylindrical article adapted to be engaged by that blank.

(17) Optionally, the maximum diameter  $D_{sub.a}$  of each aperture is less than the upper end diameter  $D_{sub.n}$  of the article.

(18) Optionally, any point on the outline of a finger aperture is disposed outside an area defined by a radius equal to half of the maximum diameter ( $\frac{1}{2}D_{sub.c}$ ) around the centre of any one of the apertures which surround the finger aperture.

(19) Within the scope of this application it is envisaged or intended that the various aspects, embodiments, examples, features and alternatives set out in the preceding paragraphs, in the claims and/or in the following description and drawings may be considered or taken independently or in any combination thereof.

(20) Features or elements described in connection with, or relation to, one embodiment are applicable to all embodiments unless there is an incompatibility of features. One or more features or elements from one embodiment may be incorporated into, or combined with, any of the other embodiments disclosed herein, said features or elements extracted from said one embodiment may be included in addition to, or in replacement of one or more features or elements of said other embodiment.

(21) A feature, or combination of features, of an embodiment disclosed herein may be extracted in isolation from other features of that embodiment. Alternatively, a feature, or combination of features, of an embodiment may be omitted from that embodiment.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) Exemplary embodiments of the invention will now be described with reference to the accompanying drawings, in which:

(2) FIG. 1 is a plan view from above of a plurality of connected blanks, each blank for forming a carrier according to a first embodiment;

(3) FIG. 2 is a plan view from above of one the blanks of FIG. 1;

(4) FIG. 2A is an enlarged view of a portion of the blank of FIG. 2;

(5) FIG. 3 is a perspective view of a carton formed from the blank of FIG. 2; and

(6) FIG. 4 is a perspective view of a primary product container for use with the carrier of the first embodiment.

### DETAILED DESCRIPTION OF EMBODIMENTS

(7) Detailed descriptions of specific embodiments of the package, blanks and cartons are disclosed herein. It will be understood that the disclosed embodiments are merely examples of the way in which certain aspects of the invention can be implemented and do not represent an exhaustive list of all of the ways the invention may be embodied. As used herein, the word “exemplary” is used expansively to refer to embodiments that serve as illustrations, specimens, models, or patterns. Indeed, it will be understood that the packages, blanks and cartons described herein may be embodied in various and alternative forms. The Figures are not necessarily to scale and some features may be exaggerated or minimised to show details of particular components. Well-known components, materials or methods are not necessarily described in great detail in order to avoid obscuring the present disclosure. Any specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the invention.

(8) Referring to FIG. 1, there is shown a plan view of a plurality of connected blanks **10a**, **10b**, **10c**, **10d** capable of forming a carton or carrier **90**, as shown in FIG. 3, for containing and carrying a group of primary products such as, but not limited to, bottles or cans, hereinafter referred to as articles B, as shown in FIG. 2. The blanks **10a**, **10b**, **10c**, **10d** form a secondary package for packaging at least one primary product container or package.

(9) In the embodiments detailed herein, the terms “carton” and “carrier” refer, for the non-limiting purpose of illustrating the various features of the invention, to a container for engaging and

carrying articles, such as primary product containers. It is contemplated that the teachings of the invention can be applied to various product containers, which may or may not be tapered and/or cylindrical. Exemplary containers include bottles (for example metallic, glass or plastics bottles), cans (for example aluminium cans), tins, pouches, packets and the like.

(10) The blanks **10a**, **10b**, **10c**, **10d** are formed from a sheet of suitable substrate. It is to be understood that, as used herein, the term “suitable substrate” includes all manner of foldable sheet material such as paperboard, corrugated board, cardboard, plastic, combinations thereof, and the like. It should be recognised that one or other numbers of blanks may be employed, where suitable, for example, to provide the carrier structure described in more detail below.

(11) The packaging structures or cartons described herein may be formed from a sheet material such as paperboard, which may be made of or coated with materials to increase its strength. An example of such a sheet material is tear-resistant NATRALOCK® paperboard made by WestRock Company. It should be noted that the tear resistant materials may be provided by more than one layer, to help improve the tear-resistance of the package. Typically, one surface of the sheet material may have different characteristics to the other surface. For example, the surface of the sheet material that faces outwardly from a finished package may be particularly smooth and may have a coating such as a clay coating or other surface treatment to provide good printability. The surface of the sheet material that faces inwardly may, on the other hand, be provided with a coating, a layer, a treatment or be otherwise prepared to provide properties such as one or more of tear-resistance, good glue-ability, heat sealability, or other desired functional properties.

(12) In the illustrated embodiments, the blanks **10a**, **10b**, **10c**, **10d** are configured to form a carton or carrier **90** for packaging an exemplary arrangement of exemplary articles B. In the embodiment illustrated, each blank **10a**, **10b**, **10c**, **10d** forms a package having an arrangement in the form of a 2×3 matrix or array; in the illustrated embodiment two rows of three articles are provided, and the articles B are beverage cans. The beverage cans may be 12 oz (355 ml) cans of the sleek or slim design; that is to say the articles B are substantially of the same diameter over their entire height. An exemplary article B is illustrated in FIG. 4, the article B has a maximum diameter or lateral dimension D.sub.c (the diameter D.sub.c may be about 2.25 inches or about 58 mm). The article B comprises an upper portion or top closure T which has a diameter or lateral dimension D.sub.n (the diameter D.sub.n may be about 2.125 inches or 54.8 mm). The neck N may provide an outwardly projecting flange, that is to say it may comprise an undercut for engaging with the carrier. A top closure may be attached to the side wall of the article B to form a seam or “chime” C which provides the flange F. In embodiments of the inventions the variation in diameter between the top closure T and the main body M of the article B is less than 7 mm, may be less than 5 mm and optionally is less than 4 mm.

(13) As used herein the terms “sleek” or “slim” refer to article which have little or no variation in their lateral dimension between the top closure T which engages with the carrier **90** and the main body M of the article B. The articles are substantially parallel sided or of substantially uniform diameter.

(14) Referring to FIG. 1, there is shown a plurality of blanks **10a**, **10b**, **10c**, **10d**, in the illustrated embodiment there are four blanks **10a**, **10b**, **10c**, **10d** arranged in a 2×2 matrix or array. Each blank **10a**, **10b**, **10c**, **10d** is connected to at least two adjacently disposed blanks **10a**, **10b**, **10c**, **10d** by a frangible or severable connection **42**. Each blank **10a**, **10b**, **10c**, **10d** comprises a main panel **12** for forming a top wall or engaging panel of a carrier **90** (see FIG. 3).

(15) Each of the main panels **12** includes at least one article retention structure RT1, RT2, RT3, RT4, RT5, RT6. In the embodiment of FIG. 1 each main panel **12** comprises a plurality of article retention structures RT1, RT2, RT3, RT4, RT5, RT6, specifically six article retention structures RT1, RT2, RT3, RT4, RT5, RT6 arranged in a 2×3 matrix or array. In other embodiments, alternative arrangements may be employed, for example but not limited to four article retention structures RT1, RT2, RT3, RT4 arranged in a 2×2 matrix or array; in such embodiments the

plurality of blanks **10a**, **10b**, **10c**, **10d** may include six blanks severably connected to form a 2×3 matrix or array. Another alternative arrangement may be three article retention structures **RT1**, **RT3**, **RT5** arranged in a 1×3 configuration; in such an embodiment the plurality of blanks **10a**, **10b**, **10c**, **10d** may include eight blanks severably connected to form a 2×4 matrix or array. Another alternative arrangement may be two article retention structures **RT1**, **RT3**; or **RT1**, **RT2** arranged in a 1×2 configuration; in such an embodiment the plurality of blanks **10a**, **10b**, **10c**, **10d** may include twelve blanks severably connected to form a 2×6 or 3×4 matrix or array. Still another alternative arrangement may be four article retention structures **RT1**, **RT2**, **RT1**, **RT2** arranged in a 1×4 configuration; in such an embodiment the plurality of blanks **10a**, **10b**, **10c**, **10d** may include six blanks severably connected to form a 1×6 configuration. Still further alternative arrangement may be eight article retention structures **RT1**, **RT2**, **RT3**, **RT4**, **RT5**, **RT6**, **RT1**, **RT2** arranged in a 2×4 configuration; in such an embodiment the plurality of blanks **10a**, **10b**, **10c**, **10d** may include three blanks severably connected to form a 1×3 configuration.

(16) Each of the article retention structures **RT1**, **RT2**, **RT3**, **RT4**, **RT5**, **RT6** is substantially similar in construction and will therefore be described in detail with reference to a first article retention structure **RT1** as illustrated in FIGS. 2 and 2A. FIG. 2 shows a single blank **10** which has been separated from the plurality of blanks **10a**, **10b**, **10c**, **10d**. FIG. 2a shows an enlarged portion of the blank **10**.

(17) The first article retention structure **RT1** comprises an aperture **A1**. The first aperture **A1** is an eleven-sided polygon or hendecagon. In other embodiments, other polygonal shapes may be employed.

(18) A plurality of article engaging tabs **16**, **18**, **20** are arranged about the periphery of the aperture **A1**. Each tab **16**, **18**, **20** is hinged to the main panel **12**.

(19) Each tab **16**, **18**, **20** is separated from its adjacent neighbours by a linear cutline **11**. In this way each tab **16**, **18**, **20** comprises a first side edge **19** and a second side edge **21**. Each tab **16**, **18**, **20** comprises a free end edge **23** opposing a hinged edge. The free end edges **23** form engaging edges for retaining an article B, or at least a portion thereof, within the aperture **A1**. The free end edges **23** each defines a side of the polygonal shape of the first aperture **A1**. Each of the linear cutlines **11**, which define the side edges of the tabs **16**, **18**, **20**, extend from a vertex or corner of the polygonal shape of the first aperture **A1**. The linear cutlines **11** may be substantially radially arranged with respect to a notional circle that passes through each of the vertices of the polygonal shape of the first aperture **A1**. The linear cutlines **11** comprise a first proximal end that intersects with a vertex or corner of the polygonal shape of the first aperture **A1**. The linear cutlines **11** comprise a second distal end.

(20) The plurality of article engaging tabs **16**, **18**, **20** comprises a series or set of first article engaging tabs **16**, a series or set of second article engaging tabs **18**, and a series or set of third article engaging tabs **20**.

(21) The set of first article engaging tabs **16** are located on the main panel **12** in a region in which the article engaging tabs are subject to the greatest stress or deformation when an article B is received in the first article retention structure **RT1**.

(22) A first arcuate cutline **26** is disposed proximate each of the linear cut lines **11** defining the side edges of the first article engaging tabs **16**. Each first arcuate cutline **26** is spaced apart from the second distal end of the linear cutlines **11** so as to define a connecting portion **28** or “nick” between a pair of adjacent first article engaging tabs **16**.

(23) The set of third article engaging tabs **20** are located on the main panel **12** in a region in which the article engaging tabs are subject to the least stress or deformation when an article B is received in the first article retention structure **RT1**.

(24) A third arcuate cutline **22** is disposed proximate each of the linear cut lines **11** defining the side edges of the third article engaging tabs **20**. Each third arcuate cutline **22** is spaced apart from the second distal end of the linear cutlines **11** so as to define a connecting portion **28** or “nick” between

a pair of adjacent third article engaging tabs **20**. Those connecting portions **28** or “nicks” are provided for maintaining a connection between a pair of adjacent tabs **16, 18, 20** even after an article B is inserted into the aperture **A1, A2, A3, A4, A5, A6**. The connecting portions **28** connect the respective tab **16, 18, 20** with the next adjacent tab **16, 18, 20**, thereby preventing or inhibiting the respective tab **16, 18, 20** from wobbling or rotating about the axis denoted by notional line P-P' or at least mitigating against such wobbling or rotation.

(25) The set of second article engaging tabs **18** are located on the main panel **12** so as to provide a transition between one of the first article engaging tabs **16** and one of the third article engaging tabs **20**.

(26) A second arcuate cutline **24, 25** is disposed proximate a cut line **11** separating each of the second article engaging tabs **18** from an adjacent third article engaging tab **20**.

(27) Each second arcuate cutline **24, 25** is spaced apart from the second distal end of the linear cutlines **11** so as to define a connecting portion **28** or “nick” between a respective one of the second article engaging tabs **18** and the third article engaging tab **20** adjacent to it.

(28) Each of the first and third arcuate cutlines **22, 26** is arranged symmetrically about the linear cutline with which it is associated. Each of the second arcuate cutlines **24, 25** is asymmetrically arranged about one of the linear cutlines **11**.

(29) The first, second and third arcuate cutlines **26, 24, 25, 22** provide stress relief in the main panel **12** when the first, second and third article engaging tabs **16, 18, 20** are displaced out of the plane of the main panel **12**.

(30) The first and third arcuate cutlines **26, 22** are arranged symmetrically about the respective linear cutline **11** with which they are associated. The first arcuate cutlines **26** are larger in dimension than the third arcuate cutlines **22**. The first arcuate cutlines **26** comprise a first radius of curvature, the third arcuate cutlines **22** comprise a second radius of curvature; the first radius of curvature is larger than the second radius of curvature.

(31) Each of the second cutlines **24, 25** is arranged asymmetrically about the respective linear cutline **11** with which it is associated. The second cutlines **24, 25** comprise a first portion and a second portion contiguously arranged with each other. The first portion is disposed proximate the second article engaging tab **18** and the second portion is disposed proximate a third article engaging tab **20**. The first portion of each of the second cutlines **24, 25** comprises a first radius of curvature, the second portion of each of the second cutlines **24, 25** comprises a second radius of curvature; the first radius of curvature is larger than the second radius of curvature.

(32) The second cutlines **24, 25** may be considered to comprise one half of a first arcuate cutline **26** and one half of a third arcuate cutline **22** contiguously arranged with each other.

(33) The second cutlines **24, 25** form asymmetrical ‘C’-shaped cuts, whereas the first and third cutlines **26, 22** form symmetrical ‘C’-shaped cuts. The second cutlines **24, 25** are employed at the boundary between a first area of the main panel **12** that is subject to higher stress upon displacement of the first article engaging tabs **16** and a second area of the main panel **12** that is subject to lower stress upon displacement of the third article engaging tabs **20**. The higher stress area of the main panel **12** occurs where the first tabs **16** are located as these first tabs **16** undergo higher bending stress, when an article B is inserted into the respective aperture **A1, A2, A3, A4, A5, A6** than the third tabs **20** disposed in the lower stress area of the main panel **12**.

(34) Each of the third article engaging tabs **20** is defined in part by a second linear cutline **17** provided in the main panel **12**.

(35) Each second linear cutline **17** is disposed between a pair of adjacent arcuate cutlines **26, 24, 25, 22** in a spaced apart relationship with each of the pair of adjacent arcuate cutlines **26, 24, 25, 22**. Each of the third article engaging tabs **20** adjacent to one of the second article engaging tabs **18** comprises a second linear cutline **17** disposed between a first arcuate cutline **26** and second arcuate cutlines **24, 25** in a spaced apart relationship with respect to both the first arcuate cutline **26** and the second arcuate cutlines **24, 25**. The remaining third article engaging tabs **20** comprise a second

linear cutline **17** disposed between a pair of adjacent first arcuate cutlines **26** in a spaced apart relationship with each of the pair of adjacent first arcuate cutlines **26**.

(36) The second linear cutline **17** facilitates folding of each of the third article engaging tabs **20** with respect to the main panel **12**.

(37) The second linear cutline **17** defines at least in part a straight or linear fold line **17** by which each of the third article engaging tabs **20** is hinged to the main panel **12**.

(38) In the illustrated embodiment the first article retention structure RT1 comprises eleven tabs **16**, **18**, **20** arranged about the periphery of the aperture A1.

(39) Optionally, the plurality of article engaging tabs **16**, **18**, **20** may vary in dimension according to their location on the main panel **12**. The first article engaging tabs **16** may have a first width, the second article engaging tabs **18** may have a second width and the third article engaging tabs **20** may have a third width. The third width may be greater than the second width which in turn may be greater than the first width. In this way the free end edge **23**, which forms an engaging edge E1, E2, of the first tabs **16** is smaller in dimension than the free end edge **23** or engaging edge of the second or third tabs **18**, **20**.

(40) In the illustrated embodiment, the article engaging tabs **16**, **18**, **20** located in the region of the main panel **12** and subject to the greatest stress or deformation when an article B is received in the article retention structure RT1, RT2, RT3, RT4, RT5, RT6 are smaller in dimension than the article engaging tabs **16**, **18**, **20** located in the region of the main panel **12** subject to the least stress or deformation.

(41) The main panel **12** may optionally comprise a handle structure. The handle structure may comprise a pair of handle apertures H. Each of the pair of handle apertures H is struck from the main panel **12**. One of the pair of handle apertures H is located in a region disposed centrally between a first pair of article retention structures RT1, RT2 and a second pair of article retention structures RT3, RT4. Said one of the pair of handle apertures H comprises a periphery which periphery is spaced from the centres of each of the apertures A1, A2, A3, A4 by a distance equal to or greater than the maximum diameter D.sub.c of the articles B. The other one of the pair of handle apertures H is located in a region disposed centrally between the and a second pair of article retention structures RT3, RT4 and a third pair of article retention structures RT5, RT6. Said other one of the pair of handle apertures H comprises a periphery which periphery is spaced from the centres of each of the apertures A3, A4, A5, A6 by a distance equal to or greater than the maximum diameter D.sub.c of the articles B.

(42) Optionally, the side and/or end edges of the main panel **12** may be arranged in a curvilinear or undulating shape.

(43) The corners of the main panel **12** are rounded, the corners define an arc having a radius of curvature R.sub.b. The arc may be centred on the centre of the aperture A1, A2, A5, A6 of the endmost article retention structures RT1, RT2, RT5, RT6.

(44) The main panel **12** includes at least a paperboard substrate and a tear resistant layer laminated together. It optionally includes an adhesive layer between the paperboard substrate and the tear resistant layer. The material of the paperboard substrate may be selected from any conventional paperboard, for example, ranging in weight upwardly from about 10 pt., preferably from about 11 pt. to about 14 pt. An example of such a substrate is a 12-point SBS board or CNK board manufactured by WestRock Company. The paperboard substrate may be a bleached or unbleached board. The board may be coated on at least one side, optionally the side opposite the lamination, with a conventional coating selected for compatibility with the printing method and board composition.

(45) The tear resistant layer may be disposed over the uncoated side of the paperboard substrate and may be formed of polymeric material and secured to the substrate. The tear resistant layer imparts toughness to the laminate structure. Suitable tear resistant materials may include, but not be limited to, tear resistant laminated sheet material, e.g., NATRALOCK®, which may include a layer



of an n-axially oriented film, e.g. MYLAR®, which is a bi-axially oriented polyester, oriented nylon, cross-laminated polyolefin or high density polyolefin. The orientation and cross-laminated structure of these materials contribute to the tear resistant characteristic. Also, tear resistance may be attributed to the chemical nature of the tear resistant material such as extruded metallocene-catalyzed polyethylene (mPE).

(46) Alternatively, the tear resistant layer may be a layer of linear low-density polyethylene (LLDPE). In embodiments where linear low-density polyethylene (LLDPE) or mPE is used, it is not necessary to incorporate an adhesive layer. Other suitable materials having a high level of tear resistance may also be used.

(47) The adhesive layer may be formed of polyolefin material such as a low-density polyethylene (LDPE). The adhesive layer may be placed between the substrate and the tear resistant layer to secure the tear resistant layer to the substrate.

(48) The positions of the articles B with respect to the blanks **10a**, **10b**, **10c**, **10d** are indicated by notional lines CP. The notional lines CP indicate the position of the cylindrical wall of the main body of an article B shown in FIG. 4 with respect to the blanks **10a**, **10b**, **10c**, **10d**. In the embodiment of FIG. 1 the position of twenty-four articles B with respect to the plurality of blanks **10a**, **10b**, **10c**, **10d** is indicated. The articles B are disposed in four distinct groups of six articles.

(49) Each article B is aligned with one of the apertures A1, A2, A3, A4, A5, A6 of the article retention structure RT1, RT2, RT3, RT4, RTS, RT6 of one of the blanks **10a**, **10b**, **10c**, **10d**. The tubular axis of the articles B is in registry with the centre of one of the apertures A1, A2, A3, A4, A5, A6.

(50) Each article B with an article group is disposed in touching contact, or in at least close proximity, with at least two adjacent neighbours. The articles B in one group are spaced apart from the nearest adjacent articles B in the neighbouring group by a distance z. This enables each of the adjacently disposed blanks **10a**, **10b**, **10c**, **10d** to provide a sufficiently thick portion of the main panel **12** about the articles B when the plurality of blanks **10a**, **10b**, **10c**, **10d** are applied simultaneously to all of the article groups. The spacing apart of the article groups by the distance z is particularly important when packaging articles of the “sleek” or “slim” design as shown in FIG. 4 where there is little or no difference in diameter between the top end closure and/or neck with respect to the side wall of the main body M of the article B.

(51) Each blank **10a**, **10b**, **10c**, **10d** comprises a maximum width W, as shown in FIG. 2. The main panel **12** is arranged to be wider than the width of the group of articles B which it accommodates, the edge of the main panel **12** is spaced a distance y from the position of the cylindrical side wall CP of the article B, where  $y=(\frac{1}{2}z)$ .

(52) The plurality of blanks **10a**, **10b**, **10c**, **10d** comprises a maximum width 2W.

(53) Each blank **10a**, **10b**, **10c**, **10d** comprises a maximum length L, as shown in FIG. 2. The main panel **12** is arranged to be longer than the length of the group of articles B which it accommodates, the edge of the main panel **12** is spaced a distance y from the position of the cylindrical side wall CP of the article B, where  $y=(\frac{1}{2}z)$ .

(54) The plurality of blanks **10a**, **10b**, **10c**, **10d** comprises a maximum length 2L.

(55) In this way the outer perimeter of the main panel **12**, in the flat blank form, is spaced a distance equal to or greater than distance y from the group of articles B as shown in FIG. 2.

(56) The main panel **12** therefore comprises a boundary region surrounding the article group.

(57) The apertures A1, A2, A3, A4, A5, A6 have a maximum lateral dimension or diameter of D.sub.a.

(58) The centre of any one of the first, second, third, fourth, fifth and sixth apertures A1, A2, A3, A4, A5, A6 of a given blank **10a**, **10b**, **10c**, **10d** is spaced from the centre of an adjacent one of the apertures A1, A2, A3, A4, A5, A6 of said blank **10a**, **10b**, **10c**, **10d** by a dimension D.sub.1, dimension D.sub.1 may be substantially equal to D.sub.c, (D.sub.1=D.sub.c). The cylindrical axes of the articles B of a given article group may be similarly spaced apart.

(59) The centre of a first aperture **A1** of a first blank **10a** is spaced from the centre of an adjacent aperture **A2**, of a second blank **10b**, by a dimension  $D_{sub.3}$ , dimension  $D_{sub.3}$  may be substantially equal to the sum of dimension  $D_{sub.c}$  and dimension  $z$ ; ( $D_{sub.3}=D_{sub.c}+z$ ). The cylindrical axes of the respective articles **B** may be similarly spaced apart.

(60) The centre of a first aperture **A1** of a first blank **10a** is spaced from the centre of an adjacent aperture **A5**, of a third blank **10c**, by a dimension  $D_{sub.2}$ , dimension  $D_{sub.2}$ , dimension  $D_{sub.2}$  may be substantially equal to the sum of dimension  $D_{sub.c}$  and dimension  $z$ ; ( $D_{sub.2}=D_{sub.c}+z$ ). The cylindrical axes of the respective articles **B** may be similarly spaced apart.

(61) The cutlines **26** of the first retention structure **RT1** are arranged to be spaced at least a distance  $x$  from the cutlines **26** of the third retention structure **RT1**. The cutlines **26** of each retention structure **RT1**, **RT2**, **RT3**, **RT4**, **RT5**, **RT6** are arranged to be spaced at least a distance  $x$  from the cutlines **26** of the an adjacent, retention structure **RT1**, **RT2**, **RT3**, **RT4**, **RT5**, **RT6**.

(62) The distance between the centre of one of the apertures **A1**, **A2**, **A3**, **A4**, **A5**, **A6** and the outer end of each one of the tabs **16**, **18**, **20** surrounding the said aperture **A1**, **A2**, **A3**, **A4**, **A5**, **A6** is given by dimension  $R_{sub.s}$ . The outer end of each one of the tabs **16**, **18**, **20** may be defined by the distal end of the radial cutlines **11**. The distance  $R_{sub.s}$  is less than or equal to half the maximum diameter  $D_{sub.c}$  of the articles **B**, ( $R_{sub.s} \leq \frac{1}{2}D_{sub.c}$ ). The distance  $R_{sub.s}$  is greater than half the maximum diameter  $D_{sub.a}$  of the apertures **A1**, **A2**, **A3**, **A4**, **A5**, **A6**, ( $R_{sub.s} > \frac{1}{2}D_{sub.a}$ ).

(63) The article retention structure **RT1**, **RT2**, **RT3**, **RT4**, **RT5**, **RT6** each define an opening which is formed in part from a respective one of the apertures **A1**, **A2**, **A3**, **A4**, **A5**, **A6** and from the plurality of article engaging tabs **16**, **18**, **20** associated with each aperture **A1**, **A2**, **A3**, **A4**, **A5**, **A6**. The opening comprise a diameter or maximum lateral dimension which may be less than the maximum diameter  $D_{sub.c}$  of the article **B** to be received therein.

(64) Referring again to FIG. 1 each of the corners of the blanks **10a**, **10b**, **10c**, **10d** may be rounded in shape. The rounded corners may be defined by a corner radius  $R_{sub.b}$ . The corner radius  $R_{sub.b}$  is greater than half the maximum diameter  $D_{sub.c}$  of the articles **B** ( $R_{sub.b} > \frac{1}{2}D_{sub.c}$ ). In some embodiments, the corner radius  $R_{sub.b}$  is  $\frac{1}{8}$  inches (3.175 mm) larger than half the maximum diameter  $D_{sub.c}$  of the articles **B**, in other embodiments the corner radius  $R_{sub.b}$  is more than  $\frac{1}{8}$  inches (3.175 mm) larger than half the maximum diameter  $D_{sub.c}$  of the articles **B**.

(65) The dimensions  $D_{sub.2}$ ,  $D_{sub.3}$  may be generally equal to twice the corner radius  $R_{sub.b}$ , ( $D_{sub.2}=D_{sub.3} \approx 2R_{sub.b}$ ).

(66) The dimensions  $D_{sub.2}$ ,  $D_{sub.3}$  are greater than the dimension  $D_{sub.1}$ , ( $D_{sub.1} < D_{sub.2}$ ;  $D_{sub.1} < D_{sub.3}$ ).

(67) Each of the handle apertures **H** are arranged to be in registry with a void between four adjacently disposed articles **B**. Each of the handle apertures **H** comprises an outline defining an edge of the main panel **12**.

(68) Each of the plurality of blanks **10a**, **10b**, **10c**, **10d** is severably connected to adjacently disposed ones of the plurality of blanks **10a**, **10b**, **10c**, **10d** by a connecting bridge portion or connecting nick **42**. The adjacently disposed blanks may be in touching contact over a short linear section **40**, see FIG. 2, of the outer edge of the blanks **10a**, **10b**, **10c**, **10d**.

(69) Turning to the construction of the carrier **90** from the blank **10**, the plurality of blanks **10a**, **10b**, **10c**, **10d** may be applied to a plurality of groups of articles **B**. The plurality of blanks **10a**, **10b**, **10c**, **10d** is lowered with respect to the groups of articles **B**. Each of the article retention structures **RT1**, **RT2**, **RT3**, **RT4**, **RT5**, **RT6** of each of the plurality of blanks **10a**, **10b**, **10c**, **10d** is aligned with a respective article **B** in one of the groups of articles **B**. Portions of the articles **B** pass through the main panels **12**. The tabs **16**, **18**, **20** of each of the article retention structures **RT1**, **RT2**, **RT3**, **RT4**, **RT5**, **RT6** are folded out of the plane of the main panels **12** and engage beneath the chime **C** (which may provide a flange **F**, see FIG. 4,) of an article **B**. In this way, the tabs **16**, **18**, **20** grip or hold the article **B** and prevent or inhibit the articles **B** from unintentionally separating from the main panels **12**. The assembled carrier **90** is shown in FIG. 3.

(70) Referring in particular to FIG. 3, each of the blanks **10a**, **10b**, **10c**, **10d** form a top engaging carrier **90**, only one of which is shown, comprising a main panel **12** which comprises apertures **A1**, **A2**, **A3**, **A4**, **A5**, **A6** each receiving a portion of a respective article B. The main panel **12** further comprises an annular series of tabs **16**, **18**, **20** formed around each of the first and second apertures **A1**, **A2**, **A3**, **A4**, **A5**, **A6**. The tabs **16**, **18**, **20** of each annular series are connected to the main panel **12** such that the tabs **16**, **18**, **20** yield out of the plane of the main panel **12**, about fold lines **31**, when an article B is received in the respective aperture **A1**, **A2**, **A3**, **A4**, **A5**, **A6** so as to bear against the article B.

(71) Another optional feature of the carrier **90** is that the main panel **12** is defined by a perimeter to which no other part of the carrier **90** is connected, once each of the blanks **10a**, **10b**, **10c**, **10d** are separated from the plurality of blanks **10a**, **10b**, **10c**, **10d**. That is to say, the carrier **90** is free of connection to other panels for example, but not limited to, side or end wall panels which extend about the sides of the article group. The perimeter of the main panel **12** is therefore defined in its entirety by free, cut or unhinged edges.

(72) Another optional feature of the carrier **90** is that the main panel **12** is defined by a perimeter including convexly curved edges and concavely curved edges, wherein the radius of curvature of the convexly curved edges is substantially equal to the radius of curvature of the concavely curved edges.

(73) The present disclosure provides a carrier of the top engaging type having improved article retention structures. In particular, the retention structures comprise article engaging tabs which yield upon insertion of an article. The tabs engage with an article to hold or secure the article within a panel of the carrier. The construction of the tabs is dependent upon the location of the tabs within the panel of the carrier. Those tabs subject to the greatest stress, typically those disposed in the interior regions of the panel or in close proximity to a tab of an adjacent retention structure, are provided with a stress relieving mechanism in the form of cutlines. The cutlines are provided proximate the side edge of the tabs and interrupt or define a fold line between the tab and the panel from which it is struck or formed.

(74) The dimension of a tab may also be dependent upon its location in the panel of the carrier. Those tabs disposed in higher stress areas may be smaller in width than tabs subject to lesser stress.

(75) The carrier is formed from a blank which is adapted to be applied to a group of articles simultaneously with at least one further blank which is similarly arranged. The blank and further blanks being severably from each other upon application to their respective groups of articles. The blank comprises a projecting edge forming a border or brim which surrounds the group of articles which is accommodates.

(76) It will be recognised that as used herein, directional references such as “top”, “bottom”, “base”, “front”, “back”, “end”, “side”, “inner”, “outer”, “upper” and “lower” do not necessarily limit the respective panels to such orientation, but may merely serve to distinguish these panels from one another.

(77) As used herein, the terms “hinged connection” and “fold line” refer to all manner of lines that define hinge features of the blank, facilitate folding portions of the blank with respect to one another, or otherwise indicate optimal panel folding locations for the blank. Any reference to “hinged connection” should not be construed as necessarily referring to a single fold line only; indeed a hinged connection can be formed from two or more fold lines wherein each of the two or more fold lines may be either straight/linear or curved/curvilinear in shape. When linear fold lines form a hinged connection, they may be disposed parallel with each other or be slightly angled with respect to each other. When curvilinear fold lines form a hinged connection, they may intersect each other to define a shaped panel within the area surrounded by the curvilinear fold lines. A typical example of such a hinged connection may comprise a pair of arched or arcuate fold lines intersecting at two points such that they define an elliptical panel therebetween. A hinged connection may be formed from one or more linear fold lines and one or more curvilinear fold

lines. A typical example of such a hinged connection may comprise a combination of a linear fold line and an arched or arcuate fold line which intersect at two points such that they define a half moon-shaped panel therebetween.

(78) As used herein, the term “fold line” may refer to one of the following: a scored line, an embossed line, a debossed line, a line of perforations, a line of short slits, a line of half-cuts, a single half-cut, an interrupted cutline, a line of aligned slits, a line of scores and any combination of the aforesaid options.

(79) It should be understood that hinged connections and fold lines can each include elements that are formed in the substrate of the blank including perforations, a line of perforations, a line of short slits, a line of half-cuts, a single half-cut, a cutline, an interrupted cutline, slits, scores, any combination thereof, and the like. The elements can be dimensioned and arranged to provide the desired functionality. For example, a line of perforations can be dimensioned or designed with degrees of weakness to define a fold line and/or a severance line. The line of perforations can be designed to facilitate folding and resist breaking, to facilitate folding and facilitate breaking with more effort, or to facilitate breaking with little effort.

(80) The phrase “in registry with” as used herein refers to the alignment of two or more elements in an erected carton, such as an aperture formed in a first of two overlapping panels and a second aperture formed in a second of two overlapping panels. Those elements in registry with each other may be aligned with each other in the direction of the thickness of the overlapping panels. For example, when an aperture in a first panel is “in registry with” a second aperture in a second panel that is placed in an overlapping arrangement with the first panel, an edge of the aperture may extend along at least a portion of an edge of the second aperture and may be aligned, in the direction of the thickness of the first and second panels, with the second aperture.

## Claims

1. A set of connected blanks detachably connected together in a matrix fashion wherein each blank comprises a plurality of apertures, the set of connected blanks comprising: a first blank and a second blank detachably connected together along respective edges, wherein (i) a first distance (D.sub.1) is a distance between respective centers of two adjacent ones of the apertures of the first blank, (ii) a distance (Rs) is a distance between the center of one of the two adjacent apertures and an outer end of an engaging tab arranged along a perimeter of the one of the two adjacent apertures, and (iii) a spaced distance (x) is a distance between respective cutlines at the outer ends of two adjacent engaging tabs, each one of the two adjacent engaging tabs belonging to a respective one of the two adjacent apertures, wherein the first distance (D1) is substantially equal to a sum of twice the distance (Rs) and the spaced distance (x); and (iv) a second distance (D.sub.2) is a distance between a center of a corner aperture of the first blank and a center of an adjacent corner aperture of the second blank, (v) a boundary distance (z) is a distance between the outer end of an engaging tab of the corner aperture of the first blank that is nearest the connected edge of the first blank and the outer end of an engaging tab of the adjacent corner aperture of the second blank that is nearest the connected edge of the second blank, wherein the second distance (D2) is substantially equal to sum of twice the distance Rs and the boundary distance (z) and wherein the spaced distance (x) is less than the boundary distance (z).

2. The set of claim 1 wherein the first distance (D.sub.1) is substantially equal to a maximum diameter (D.sub.c) of a generally cylindrical article adapted to be engaged by the first blank.

3. The set of claim 1 wherein the second distance (D.sub.2) is generally equal to a third distance (D.sub.3) between the center of the corner aperture of the first blank and a center of an adjacent corner aperture of a third blank.

4. The set of claim 3 wherein each of the second and third distances (D.sub.2, D.sub.3) is generally equal to a difference between the first distance (D.sub.1) and the spaced distance (x) added to the

boundary distance (z).

5. The set of claim 3, wherein the second blank is detachably connected to the first blank along a first edge thereof, and wherein the third blank is detachably connected to the first blank along a second edge thereof, the second edge of the first blank being adjacent to the first edge of the first blank.

6. The set of claim 1, wherein each blank of the set is for forming a top engaging carrier applicable to articles of a slim or sleek design, such articles comprising a top end closure, a neck, and a main body, in which there is little or no difference in diameter or width between the top end closure or the neck and a side wall of the main body of the articles.

7. A set of connected blanks detachably connected together wherein each of the blanks comprises a plurality of article retention structures each comprising an aperture, the set of connected blanks comprising: a first blank and a second blank detachably connected together along respective connected edges thereof, wherein each of the first blank and the second blank comprises a boundary region adjacent its connected edge, wherein the boundary region extends a first distance (y) measured between a point corresponding to an outer end of one of the plurality of article retention structures nearest the connected edge and a point on the connected edge, and wherein the first distance (y) is greater than a second distance (x) measured between respective cut lines of two adjacent article retention structures among the plurality of article retention structures of one of the first blank and the second blank.

8. The set of claim 7 wherein each of the first blank and the second blank comprises a main panel defined by a perimeter including convexly curved edges and concavely curved edges.

9. The set of claim 8 wherein a radius of curvature of the convexly curved edges is substantially equal to a radius of curvature of the concavely curved edges.

10. The set of claim 8 wherein the first and the second blanks are connected together at their respective convexly curved edges.

11. The set of claim 8 wherein the first and the second blanks are frangibly connected together at their respective convexly curved edges such that each of the first blank and the second blank is applicable to a plurality of groups of articles simultaneously.

12. The set of claim 8 wherein the first and the second blanks are spaced apart from each other at their respective concavely curved edges.

13. The set of claim 8 wherein each of the main panels of the first blank and the second blank comprises a retention structure (RT) defining an opening formed by a respective one of the plurality of apertures.

14. The set of claim 13 wherein cutlines of the RT of the first blank are spaced apart from the cutlines of the RT of the second blank by at least a boundary distance z.

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